

Modern Computing

- Von Neumann Architecture
 - Input devices
 - Output devices
 - Storage
 - Processing unit

Input Devices



Output Devices



Monitor



Printer



Speakers



Head Set



Projector

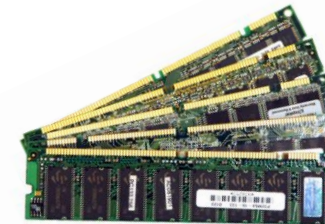


Plotter

Main Memory and Secondary Memory



hold information
permanently
for
future use



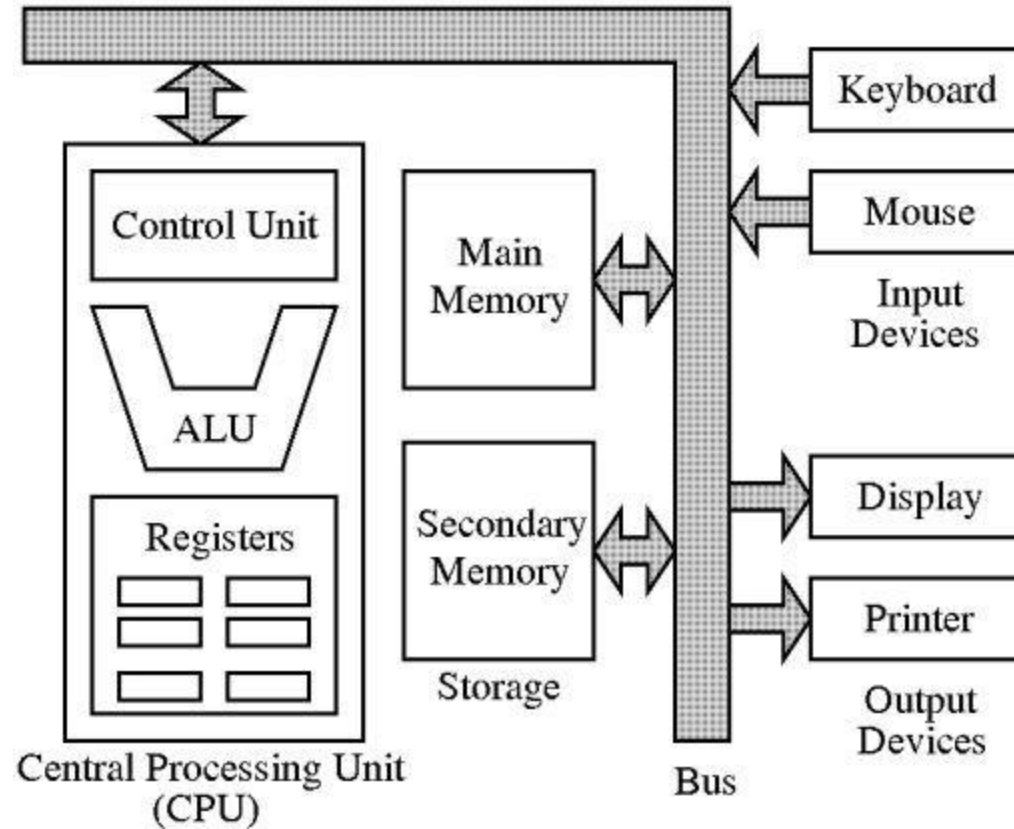
Content of memory
Temporary

Memory Access Time

Access time is the amount of time the processor to read data, instructions, and information from memory



Von Neumann Architecture



Von Neumann Architecture

There are 3 major components in a computer are put together by buses:

A. Memory: Stores and retrieves instructions and data.

B. Processor: consists of two separate components:

1) The control unit: Repeats the following 3 tasks repeatedly

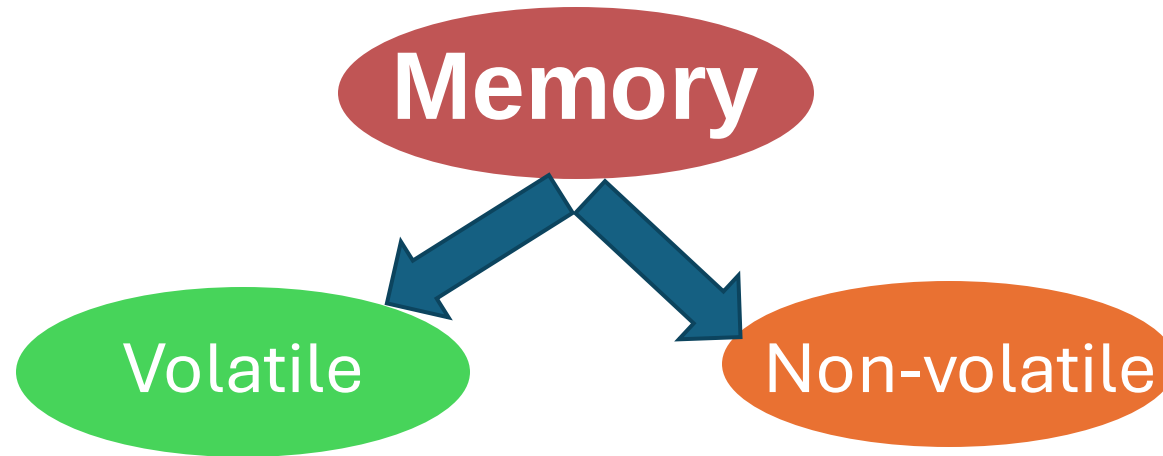
Fetches an instruction from memory

Decodes the instruction

Executes the instruction

2) The arithmetic/logic unit (ALU): Performs mathematical and logical operations.

Types of Memory



Example:

1. **Random Access Memory (RAM)**
2. **Cache**

Data/information lose when power is turn off. Temporary means for holding data, instruction or information currently being processed.

Data/information of memory retained although the power is turned off. Permanent means to hold data, instruction, information that is currently being processed.

Example:

1. **Read Only Memory (ROM)**
- data on most ROM chips cannot be modified.
2. **Flash Memory**
- Can be erased electronically and rewritten.
3. **CMOS (complementary metal-oxide semiconductor)**
- store a computer's startup information such as calendar, date and time.

Memory Size

Term	Abbreviation	Approximately Number of Bytes	Exact Amount of Bytes
Kilobyte	KB or K	1 thousand	1, 024
Megabyte	MB	1 million	1,048,576
Gigabyte	GB	1 billion	1,073,741,824
Terabyte	TB	1 trillion	1,099,511,627,776

Finally Data Representation in Computer

How is a letter converted to binary form and back?



Step 1

The user presses the capital letter **T** (SHIFT+T key) on the keyboard.

Step 2

An electronic signal for the capital letter **T** is sent to the system unit.



Step 3

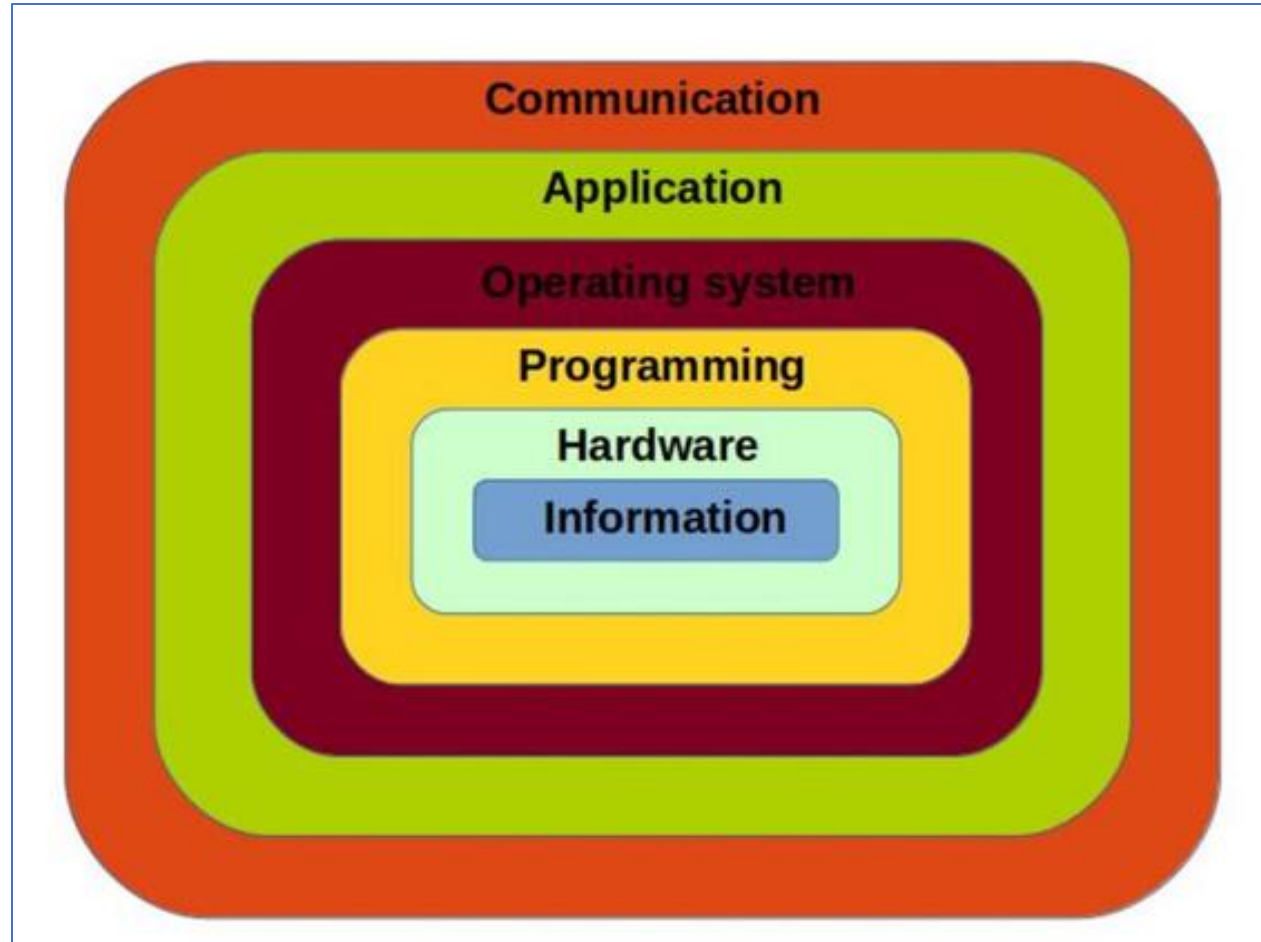
The system unit converts the scan code for the capital letter **T** to its ASCII binary code (01010100) and stores it in memory for processing.

Step 4

After processing, the binary code for the capital letter **T** is converted to an image, and displayed on the output device.



Six layers of Computing System



Where are we?

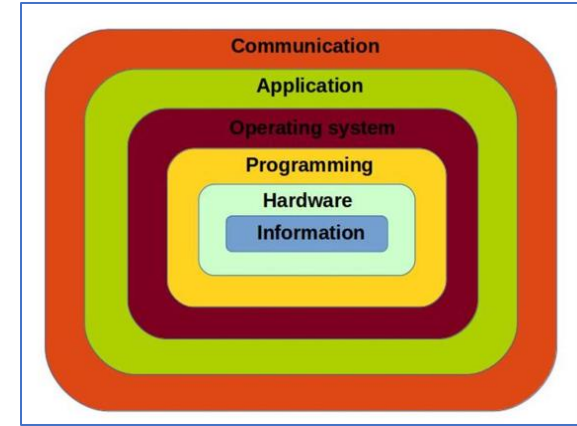
Information layer:

- Number system/Binary/Base conversions
- Data representation



Hardware layer:

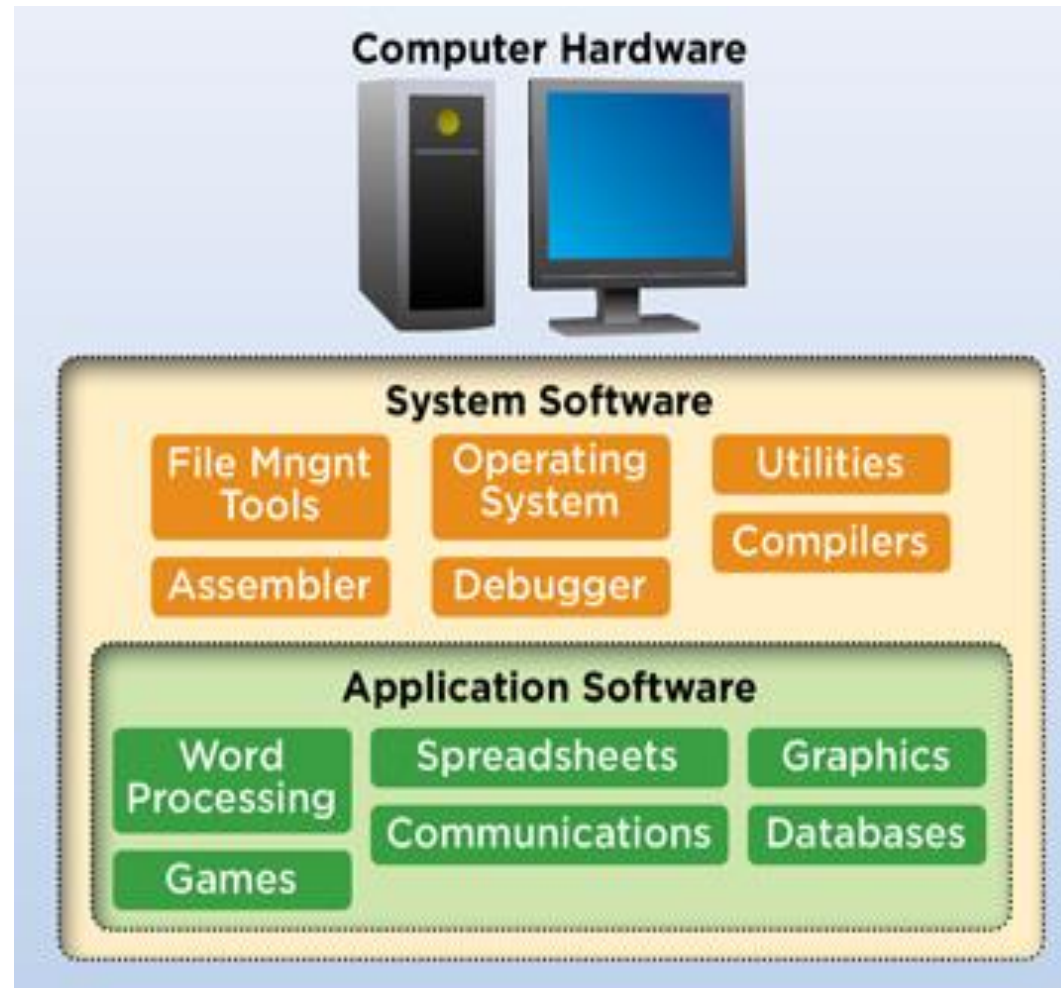
- Gates and circuits
- Computing components



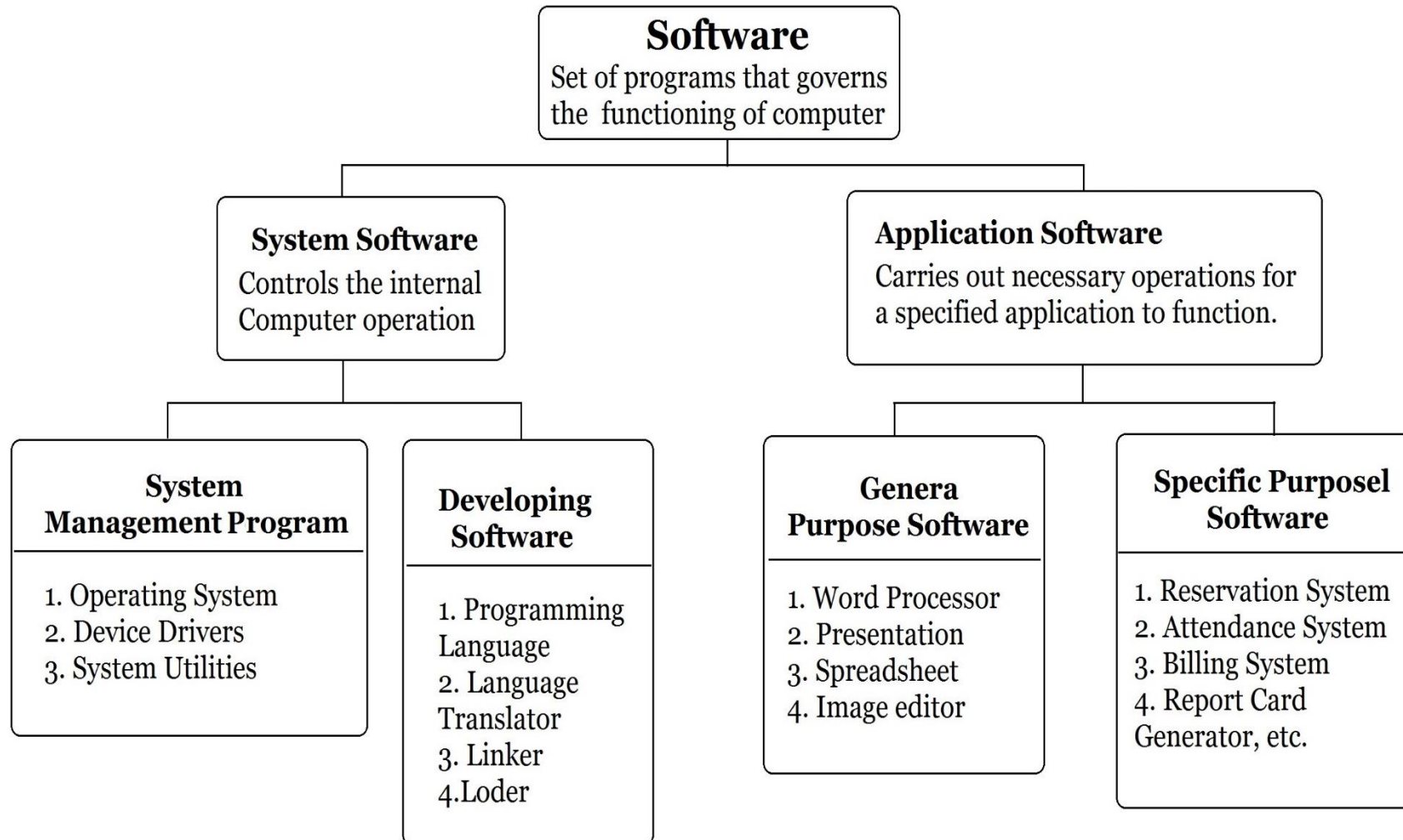
Program /Computer Software

- Series of instructions that tells the computer what to do and how to do.
- With a graphical user interface (GUI), we can interact with the software using text, graphics and visual images such as icons.
- Two categories of software:-
 - 1) System Software**
 - 2) Application Software**

Program /Computer Software



Program /Computer Software



System Software

- Consists of programs that control or maintain the operations of the computer and its devices.
- Provides the services between the user, the application software and the computer's hardware.
- Two types:
 - **Operating system**
 - **Utility program**

Operating System



Operating system – set of a programs that coordinates all the activities among computer hardware devices. Example Windows Vista, Windows XP and Mac OS X.

Utility Program

Allows a user to perform maintenance-type tasks usually related to managing a computer, its devices, or its programs.

For example utility programs to transfer digital photos to a CD or DVD.

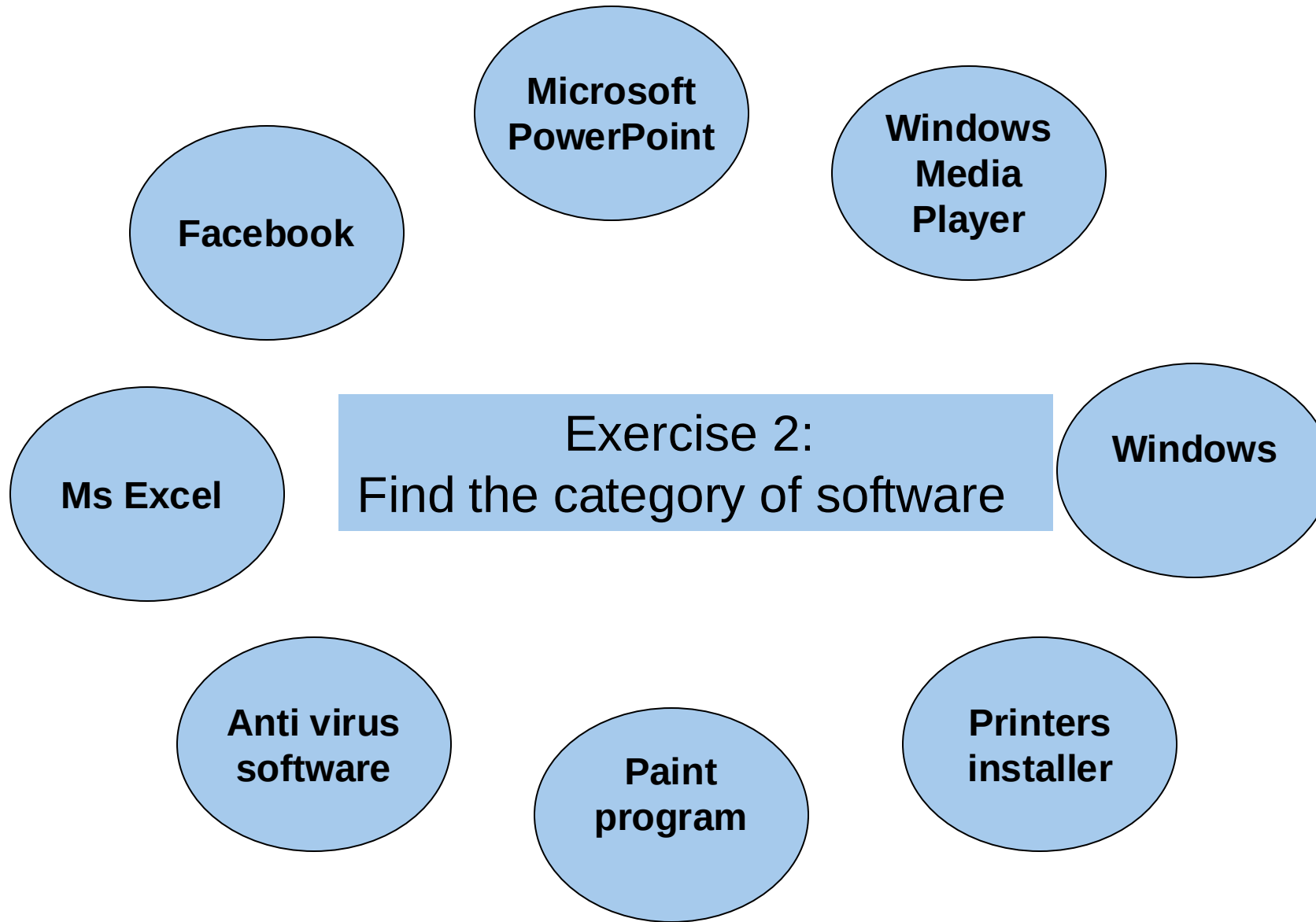
Examples of utility programs :

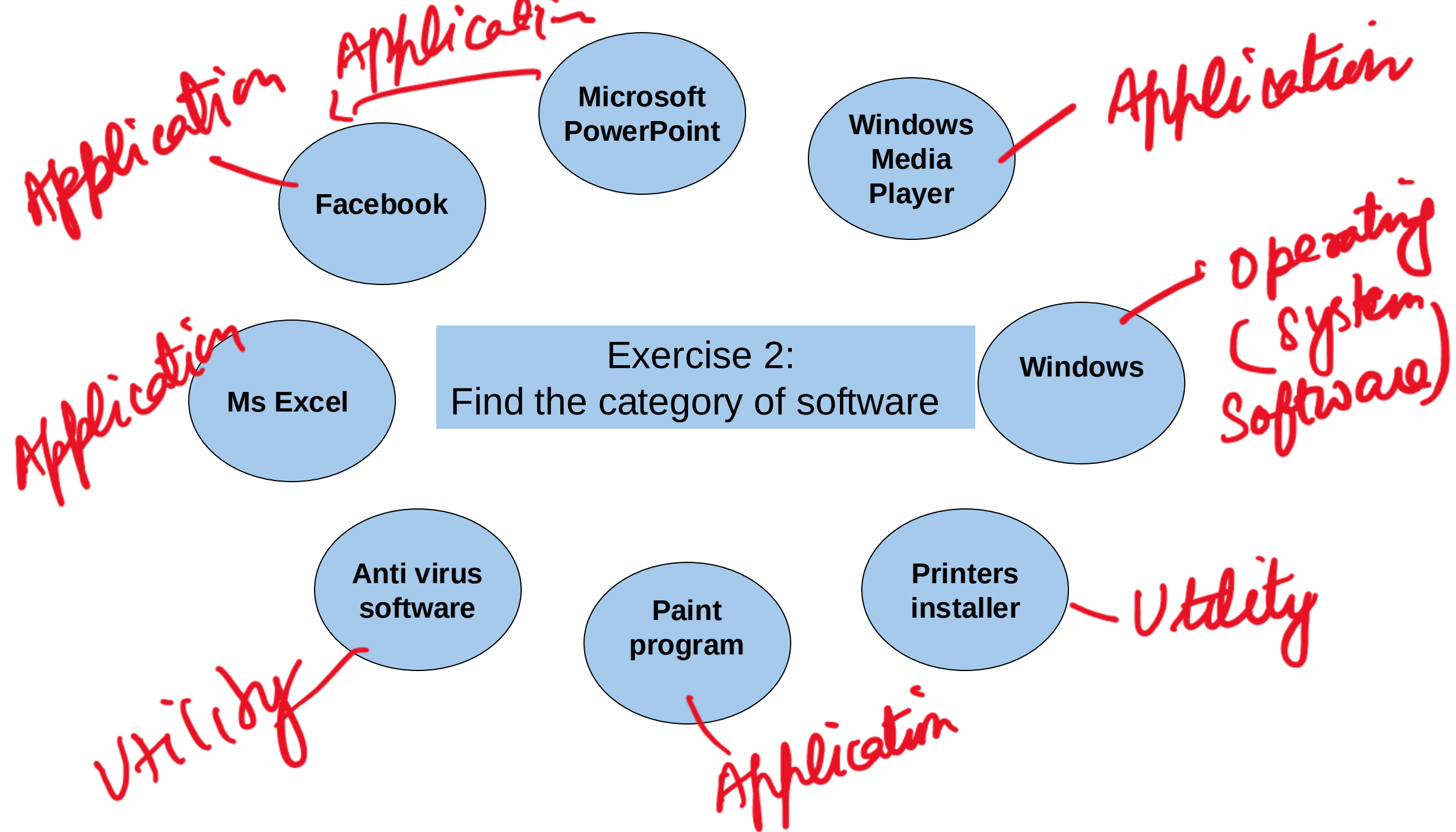
- File viewer
- Backup utility
- File compression
- Disk scanner
- Disk defragment
- Uninstaller
- Antivirus utility
- Screen saver



Application Software





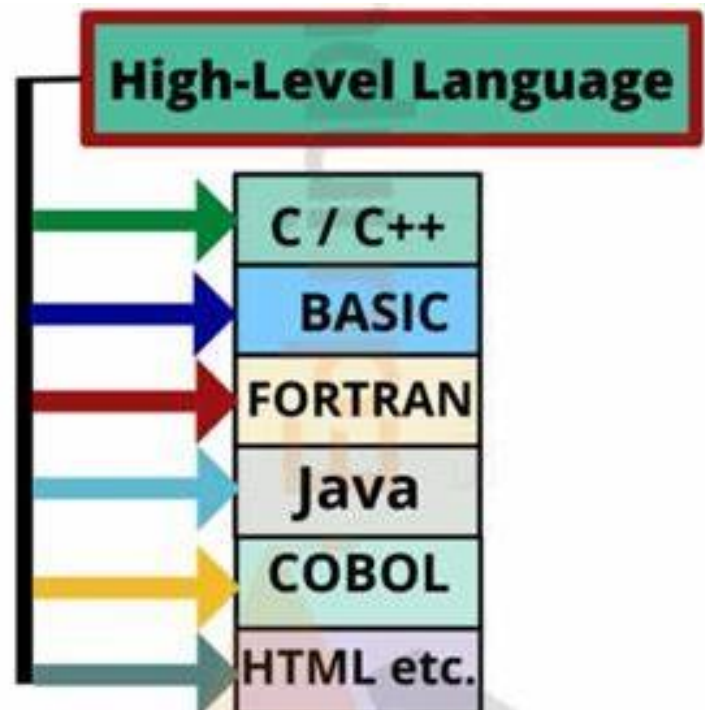


Low-level Programming languages

- Machine language:
 - Is the only language that computer can directly understand.
 - Uses a series of binary digits (1s and 0s) with a combination of numbers and letters
- Assembly language:
 - Symbolic instruction codes, meaningful abbreviations and codes.
 - **Source program** contains code to be converted to machine language
 - Uses assembler to translate to machine code

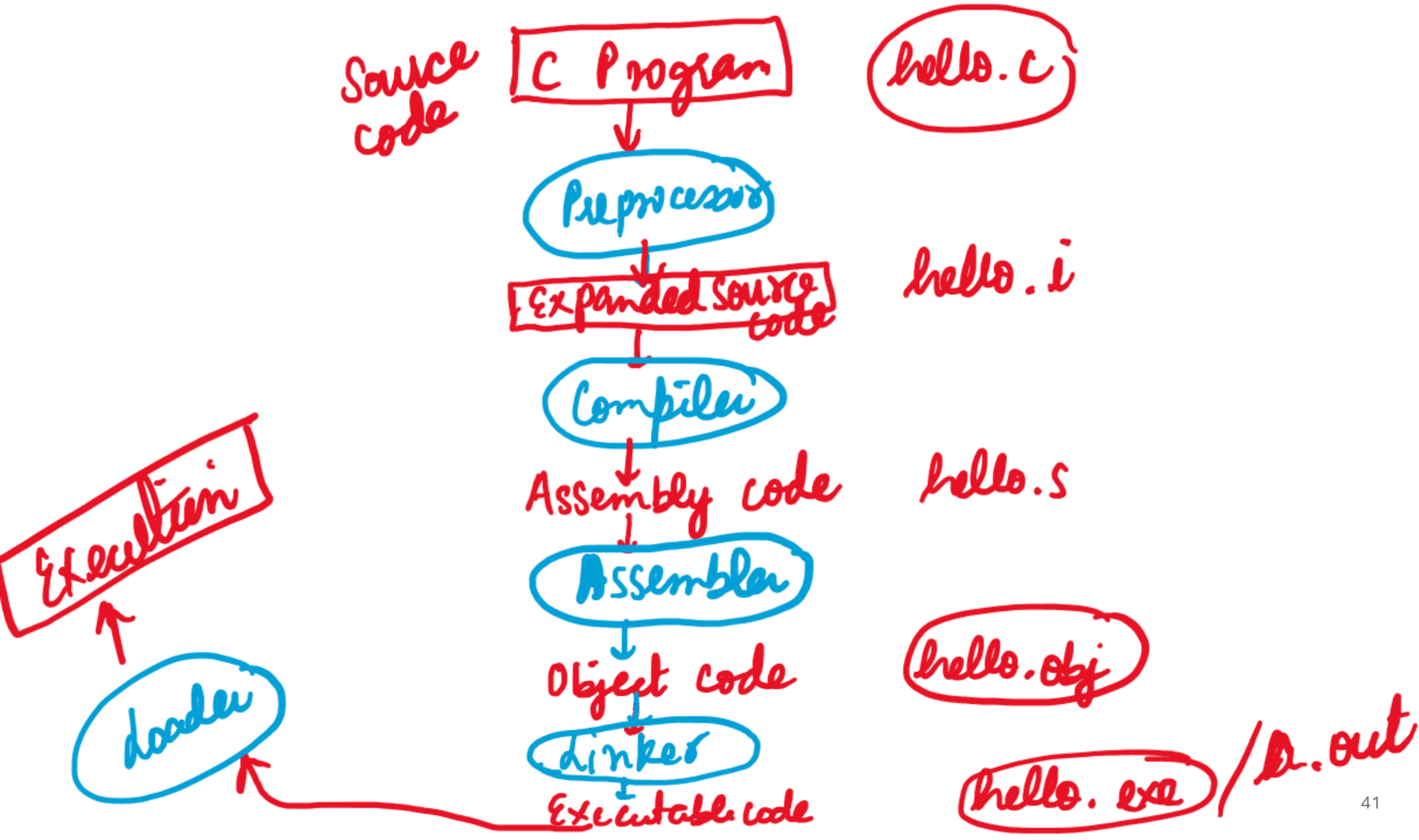
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High-level Programming language



Language Translators

A program that accept human-readable source statement (code) and produces machine-readable instruction.



Language Translators

